

Lecture 3.2: Properties of Regular Expressions

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In this lecture, you will learn some properties of regular language as well as regular expression.

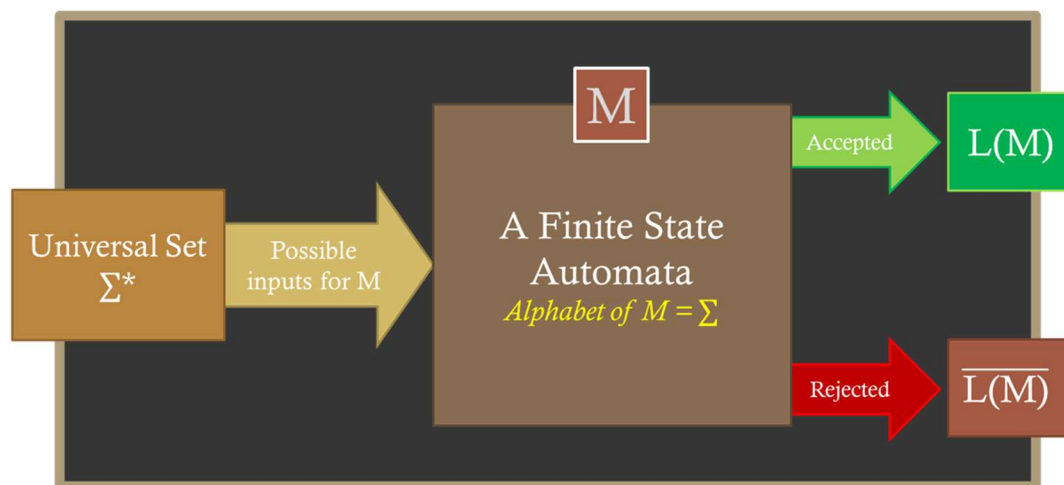
Properties of regular language

1. The **union** of any two regular languages is also a regular language. i.e. if L_1 and L_2 are two regular languages, this property follows that $L_1 \cup L_2$ must be a regular language too.

Proof: Obvious by definition. (recall the previous lecture)

2. The **complement** of a regular language is also a regular language. i.e. if L is a regular language, then this property dictates that the complement of that language, $\bar{L}$ must be a regular language too.

Proof: We can use the concept of an abstract finite state machine to prove it.



Recall the above this diagram of a finite automaton that we have seen earlier. Let, this automaton recognizes the regular language L . In the above diagram-

- M = the name of the machine
- Σ = the alphabet set for language L
- $L(M)$ = the set of strings **accepted** by the machine M
- $\bar{L}(M)$ = the set of strings that the machine M **rejects**.

Now, consider another finite state machine built on top of the above machine. Let's call this new machine **Arsenal**!

- we have built **Arsenal** following that logic that whenever M accepts something, **Arsenal** is going to reject that and vice versa.

- **Arsenal** is obviously a finite state machine because we have just added one more step after the finite state machine M i.e. whatever the original machine M produces as output, **Arsenal** will **reverse** it.
- In short, Arsenal will accept only those strings rejected by the machine M . So, the language of this **Arsenal** is nothing but $\overline{L(M)}$.

(Proved)

Prove/disprove it by yourself: Is the intersection of two regular languages regular?

Properties of Regular expressions

Let, R, M, N are some regular expressions. Now,

- OR operation is commutative but *Concatenation* is not.
So, $R + M = M + R$
However, $R \cdot M \neq M \cdot R$.
- Both OR and *Concatenation* obeys the associative law.
So, $(R + M) + N = R + (M + N)$
Also, $(RM)N = R(MN)$
- The identity of **OR** is empty language $\emptyset$. So, $\emptyset + R = R + \emptyset = R$
- The identity of ***Concatenation*** is empty string ϵ . So, $\epsilon R = R\epsilon = R$
- The annihilator of ***Concatenation*** is empty language ϕ . So, $\phi R = R\phi = \phi$
- OR follows the idempotent law. So, $R + R = R$
- Regular expressions obey the distributive law of *Concatenation*. On the contrary, it is not true for OR .
So, $(M + N)R = MR + NR$
and, $R(M + N) = RM + RN$
However, $MN + R \neq (M + R)(N + R)$
and, $R + MN \neq (R + M)(R + N)$
- **Some identities:**
 - $R^* = (R^*)^*$
 - $\emptyset^* = \epsilon$
 - $\epsilon^* = \epsilon$
 - $R^*R^* = R^*$
 - $R^*R = R^+$
 - $(M + N)^* = (M^*N^*)^* = (M^* + N^*)^*$

Is the intersection of two regular languages regular?

Proof using De Morgan's law:

Let, L & M are two regular language.

From the properties of regular language, we get,

$\Rightarrow \bar{L}$ & $\bar{M}$ are regular.

$\Rightarrow \bar{L} \cup \bar{M}$ is regular

$\Rightarrow \overline{\bar{L} \cup \bar{M}}$ is regular.

$\therefore L \cap M$ is regular. [$\because$ From Morgan's law, $\overline{\bar{x} \cup \bar{y}} = \bar{x} \cap \bar{y}$]

(Proved)
